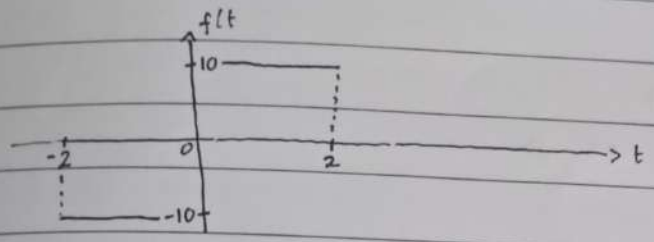
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TUGAS 7

1. $f(t) = \begin{cases} -10 & -2 < t \leq 0 \\ 10 & 0 < t < 2 \end{cases}$, dengan $f(t) = f(t+4)$



$T = 4$

$\omega = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{1}{2}\pi$

F(t) merupakan fungsi ganjil, sehingga deret fourier $= \sum_{n=1}^{\infty} b_n \sin n\omega t$

$b_n = \frac{4}{T} \int_0^{T/2} f(t) \sin n\omega t \, dt$

$= \frac{4}{4} \int_0^2 10 \sin n\frac{1}{2}\pi t \, dt$

$= -\frac{20}{n\pi} \cos \frac{1}{2} n\pi t \Big|_0^2$

$= -\frac{20}{n\pi} \cos \frac{1}{2} n\pi (2) - \left(-\frac{20}{n\pi} \cos \frac{1}{2} n\pi (0)\right)$

$= -\frac{20}{n\pi} \cos n\pi + \frac{20}{n\pi} \cos 0$

$= -\frac{20}{n\pi} \cos n\pi + \frac{20}{n\pi}$

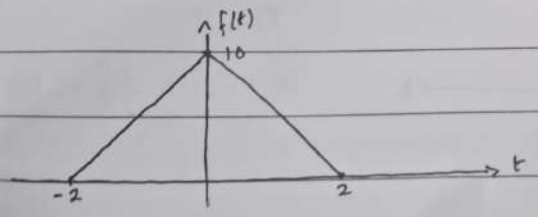
n genap $\rightarrow b_n = -\frac{20}{n\pi} \cdot 1 + \frac{20}{n\pi} = 0$

n ganjil $\rightarrow b_n = -\frac{20}{n\pi}(-1) + \frac{20}{n\pi} = \frac{40}{n\pi}$

$b_n = \begin{cases} 0 & \text{ketika n genap} \\ \frac{40}{n\pi} & \text{ketika n ganjil} \end{cases}$

Deret fourier $= \frac{40}{\pi} \sin \frac{1}{2}\pi t + \frac{40}{3\pi} \sin \frac{3}{2}\pi t + \frac{40}{5\pi} \sin \frac{5}{2}\pi t + \ldots$

2. $f(t) = \begin{cases} 10 + 5t, & -2 \leq t < 0 \\ 10 - 5t, & 0 \leq t < 2 \end{cases}$ dengan $f(t) = f(t+4)$



$T = 4$

$\omega = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{1}{2}\pi$

F(t) merupakan fungsi genap, sehingga deret fourier $= a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t$

$a_0 = \frac{2}{T} \int_{-T/2}^{0} f(t) \, dt$

$= \frac{2}{4} \int_{-2}^{0} 10 + 5t \, dt$

$= \frac{1}{2} \left[10t + \frac{5}{2} t^2\right]_{-2}^{0}$

$= \frac{1}{2} \left[0 - \left(10(-2) + \frac{5}{2}(-2)^2\right)\right]$

$= \frac{1}{2}\left[0-(-20+10)\right]$

$a_0 = 5$

$a_n = \frac{4}{T}\int_{-T/2}^{0} f(t)\cos n\omega t\, dt$

$= \frac{4}{4}\int_{-2}^{0} (10+5t)\cos \frac{1}{2}n\pi t\, dt$

hitung integral Parsial

u	dv
$10+5t$	$\cos \frac{1}{2}n\pi t$
5	$\frac{2}{n\pi}\sin \frac{1}{2}n\pi t$
0	$-\frac{4}{(n\pi)^2}\cos \frac{1}{2}n\pi t$

$a_n = \frac{20+10t}{n\pi}\sin \frac{1}{2}n\pi t + \frac{20}{(n\pi)^2}\cos \frac{1}{2}n\pi t \Big|_{-2}^{0}$

$= 0 + \frac{20}{(n\pi)^2}\cos 0 - \left(0 + \frac{20}{(n\pi)^2}\cos n\pi\right)$

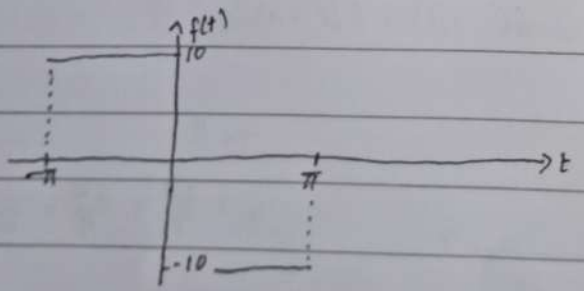
$= \frac{20}{(n\pi)^2} - \frac{20}{(n\pi)^2}\cos n\pi$

n genap $\rightarrow a_n = \frac{20}{(n\pi)^2} - \frac{20}{(n\pi)^2}\cdot(1) = 0$

n ganjil $\rightarrow a_n = \frac{20}{(n\pi)^2} - \frac{20}{(n\pi)^2}(-1) = \frac{40}{(n\pi)^2}$

Deret fourier $= 5 + \frac{40}{\pi^2}\cos \frac{1}{2}\pi t + \frac{40}{9\pi^2}\cos \frac{3}{2}\pi t + \frac{40}{25\pi^2}\cos \frac{5}{2}\pi t + \ldots$

3. $f(t) = \begin{cases} 10 & -\pi < t \le 0 \\ -10 & 0 < t < \pi \end{cases}$, dengan $f(t) = f(t+2\pi)$



$T = 2\pi$

$\omega = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$

F(t) merupakan fungsi ganjil, maka deret fourier $= \sum_{n=1}^{\infty} b_n \sin n\omega t$

$b_n = \frac{4}{T}\int_{-T/2}^{0} f(t)\sin n\omega t\, dt$

$= \frac{4}{2\pi}\int_{-\pi}^{0} 10\sin nt\, dt$

$= \frac{2}{\pi}\left[-\frac{10}{n}\cos nt\right]\Big|_{-\pi}^{0}$

$= \frac{2}{\pi}\left[-\frac{10}{n}\cos 0 - \left(-\frac{10}{n}\cos n\pi\right)\right]$

No.

Date:

$= \frac{2}{\pi}\left(-\frac{10}{n} + \frac{10}{n}\cos n\pi\right)$

n genap $\rightarrow b_n = \frac{2}{\pi}\left(-\frac{10}{n} + \frac{10}{n}(1)\right) = 0$

n ganjil $\rightarrow b_n = \frac{2}{\pi}\left(-\frac{10}{n} + \frac{10}{n}(-1)\right) = -\frac{40}{n\pi}$

Deret fourier $= -\frac{40}{\pi}\sin t - \frac{40}{3\pi}\sin 3t - \frac{40}{5\pi}\sin 5t + \ldots$